Capstone Project

Explainable AI for Autonomous Vehicle Safety: Learning from Disengagement and Crash Reports

Final Report

1. Define the Problem Statement

Autonomous vehicles are becoming more common, but safety remains one of the biggest concerns as this technology continues to develop. Even with advances in autonomous driving systems, disengagements and collisions still occur. Understanding the patterns behind these reported events is important for improving the technology and identifying areas that may need further investigation.

For my capstone project, I wanted to explore how machine learning could be used to analyze autonomous vehicle safety data. I focused on publicly available 2024 California Department of Motor Vehicles (DMV) disengagement and collision reports to see whether machine learning models could identify patterns that distinguish the two types of events.

One of the main challenges with this dataset is that there are significantly fewer collision reports than disengagement reports. This imbalance makes it difficult for machine learning models to identify collision records accurately. A model may achieve high overall accuracy while still failing to identify the less common collision reports.

My goal was to compare different machine learning models and evaluate how well they classify these reports. I also wanted to understand which available features contributed to the predictions and whether model tuning could improve performance.

This project demonstrates how machine learning can help analyze historical autonomous vehicle reports and identify patterns that may be useful for further safety research. However, the models do not predict future accidents or directly measure how safe a particular manufacturer or vehicle is.

2. Model Outcomes or Predictions

For this project, I used supervised machine learning to see whether different models could distinguish between autonomous vehicle disengagement reports and collision reports. Since there are only two possible outcomes, I treated this as a binary classification problem.

I created a target variable called Risk_Event, where 0 represents a disengagement report and 1 represents a collision report. I used manufacturer information and reporting month as the input features to see whether they could help the models identify differences between the two types of reports.

I tested several machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and Neural Networks. I also explored hyperparameter tuning to see whether I could improve the Random Forest model's performance.

The expected outcome was for each model to classify a report as either a disengagement or a collision. Since the dataset contained far fewer collision reports, I focused on precision, recall, and F1-score rather than accuracy alone. I paid particular attention to recall because I wanted to understand how many collision reports each model could correctly identify.

My goal was not just to compare model performance but also to understand what the results could tell me about the available data. However, these predictions only distinguish between historical report types. They do not predict future accidents or determine whether a particular vehicle or manufacturer is safe.

3. Data Acquisition

For this project, I used publicly available autonomous vehicle data from the California Department of Motor Vehicles (DMV). I selected the 2024 disengagement and collision reports because they provided information about events reported during autonomous vehicle testing and operation.

- 2024 Autonomous Vehicle Disengagement Reports: 1,457 records.
- 2024 Autonomous Vehicle Collision Reports: 101 records.

After combining the two datasets, I had a total of 1,558 records for my analysis.

I chose these datasets because they came from the same government agency and covered the same reporting year. Using both sources allowed me to compare two different types of reported events and explore whether machine learning could identify patterns between them.

During my initial exploration, I looked at the number of reports in each category and how the records were distributed across manufacturers and reporting months. One of the first things I noticed was the large difference between the number of disengagement and collision reports.

This was important because it showed that the dataset was imbalanced. Disengagement reports accounted for approximately 93.5% of the records, while collision reports represented only 6.5%. I considered this imbalance when selecting models and evaluating their performance.

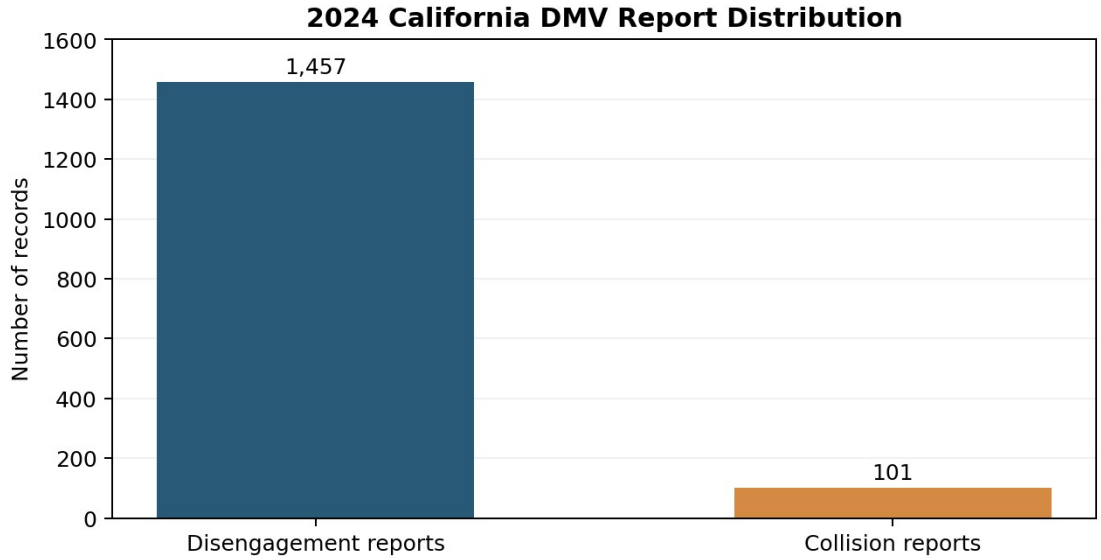


Figure 1. Distribution of the 1,558 historical DMV report records used in the analysis.

I also used visualizations to examine the distribution of report types and better understand the available data before moving on to preprocessing and modeling.

Although these datasets provided a useful starting point, they did not include all the information needed to measure vehicle safety directly. Factors such as autonomous miles

driven, driving conditions, and vehicle exposure would be important for a more complete safety analysis.

4. Data Preprocessing/Preparation

Before training the machine learning models, I spent time preparing the data to make sure it was consistent and suitable for analysis. Since I was working with two different DMV datasets, I needed to organize the information into a common format before combining them.

I started by reviewing the datasets for missing values, duplicate records, and inconsistencies in the manufacturer names. I standardized the manufacturer information so that the same manufacturer would not appear under different names. I also prepared the reporting dates and extracted the month as a numerical feature.

Next, I combined the disengagement and collision datasets into one dataset containing 1,558 records. I created a target variable called Risk_Event, where 0 represents a disengagement report and 1 represents a collision report.

For the model inputs, I selected manufacturer information and reporting month. Since manufacturer names are categorical values, I used one-hot encoding to convert them into a format that machine learning models could process. I used a preprocessing pipeline to apply the same transformations consistently during model training and evaluation.

I then divided the dataset into 80% training data and 20% testing data using a stratified split. This helped maintain approximately the same proportion of collision and disengagement reports in both sets. I also used a fixed random state of 42 to make the results reproducible.

One of the main challenges was the imbalance between the two report categories. Since collision reports were much less common, I explored class balancing techniques for selected models, including the Neural Network, to help the models pay more attention to the minority class.

For the Neural Network, I also applied feature scaling using StandardScaler as part of the modeling pipeline.

These preprocessing steps helped me prepare the data for model training and made it easier to compare the different algorithms using a consistent approach.

5. Modeling

For this project, I tested several machine learning models to see how well they could distinguish between autonomous vehicle disengagement and collision reports. I wanted to compare different approaches rather than rely on a single model.

I started with Logistic Regression as a baseline because it is relatively simple and easy to interpret. This gave me a starting point for comparing the performance of more complex models.

Next, I tested Decision Tree and Random Forest models. I selected Decision Tree because it can capture relationships in the data and is relatively easy to understand. Random Forest combines multiple decision trees, which can help improve model stability and reduce overfitting.

I also tested Gradient Boosting to see whether combining trees sequentially could improve classification performance. In addition, I explored Neural Networks to compare their results with the traditional machine learning models.

Since the dataset contained significantly fewer collision reports than disengagement reports, I experimented with class balancing techniques. For the Neural Network, I used sample weights to give more importance to collision records during training.

After evaluating the initial models, I used GridSearchCV and RandomizedSearchCV to tune the Random Forest hyperparameters. I experimented with settings such as the number of trees, maximum tree depth, and minimum samples required to split nodes or create leaf nodes.

I also explored additional Neural Network configurations to see whether adjusting the model could improve its ability to identify collision reports.

Throughout the modeling process, I used the same training and testing split to make the comparisons consistent. My goal was to find a model that could identify a reasonable number of collision reports while keeping false positives manageable.

Rather than selecting a model based only on accuracy, I compared precision, recall, and F1-score to understand the strengths and weaknesses of each approach.

6. Model Evaluation

For this project, I evaluated several supervised classification models to see how well they could distinguish between autonomous vehicle disengagement and collision reports. Since the dataset contained significantly fewer collision reports, I focused on precision, recall, and F1-score rather than accuracy alone.

I used accuracy to measure overall classification performance, precision to understand how many predicted collision reports were actually collisions, and recall to measure how many collision reports the models correctly identified. I also used F1-score to compare the balance between precision and recall.

Initial Model Comparison

Model	Accuracy	Precision	Recall	F1-score
RandomizedSearch CV Random Forest	0.8462	0.2742	0.85	0.4146
Random Forest	0.8494	0.2712	0.80	0.4051
GridSearchCV Random Forest	0.8462	0.2667	0.80	0.4000
Gradient Boosting	0.8077	0.2368	0.90	0.3750
Decision Tree	0.8077	0.2222	0.80	0.3478
Balanced Neural Network	0.7788	0.2118	0.90	0.3429
Logistic Regression	0.7692	0.1977	0.85	0.3208
Original Neural Network	0.9359	0.0000	0.00	0.0000

In my initial comparison, RandomizedSearchCV Random Forest achieved the highest F1-score of 0.4146, with 85% recall. This model correctly identified 17 out of 20 collision reports in the test set. Although the improvement over the original Random Forest was small, it provided the best balance between precision and recall among the eight configurations.

Gradient Boosting and the Balanced Neural Network achieved the highest recall of 90%, but they also produced more false positives. The original Neural Network achieved the highest accuracy of 93.59%, but it failed to identify any collision reports. This showed me why accuracy alone can be misleading when working with imbalanced datasets.

Final Model Optimization

After the initial comparison, I continued experimenting with the Random Forest and Neural Network models. I wanted to see whether further adjustments could improve their ability to identify collision reports.

Model	Accuracy	Precision	Recall	F1-score
Tuned Random Forest	0.8462	0.2667	0.80	0.4000
Optimized Neural Network	0.8045	0.2267	0.85	0.3579

The final tuned Random Forest achieved an F1-score of 0.4000 and correctly identified 16 of the 20 collision reports. The optimized Neural Network achieved 85% recall, correctly identifying 17 collision reports, but its lower precision meant it also generated more false positives.

Based on these results, the Random Forest provided a better balance between precision and recall, while the optimized Neural Network identified slightly more collision reports.

Key Findings and Limitations

One of the most important things I learned from this project was that model accuracy does not always reflect how well a model performs, especially when the dataset is imbalanced. Precision, recall, and F1-score gave me a better understanding of the trade-offs between the models.

I also found that hyperparameter tuning did not always improve performance. Some configurations performed better during cross-validation but did not produce better results on the held-out test set.

Although RandomizedSearchCV Random Forest achieved the highest F1-score in the initial comparison, the differences between several models were relatively small. Since the test set contained only 20 collision reports, the results may change with additional data.

Another limitation is that I used manufacturer and reporting month as the main input features. These features do not capture important factors such as road conditions, vehicle mileage, traffic, or driving environment.

Most importantly, the models classify historical collision and disengagement reports. They do not predict future accidents or establish whether one manufacturer is safer than another.

Recommendations and Next Steps

Based on these results, I would recommend evaluating models on imbalanced safety-report data using recall and F1-score alongside accuracy. A model with high accuracy can still miss collision records, so it is important to review the confusion matrix and the number of false positives as well.

Among the models I tested, the Random Forest approaches provided the strongest balance of precision and recall, although their precision was still limited. These results are best used to understand historical reporting patterns rather than make operational safety decisions.

As a next step, I would like to include additional information such as autonomous miles driven, road conditions, and driving environment. I would also test the approach on a different reporting year to see whether the results hold up with new data.

Conclusion

Overall, this project helped me understand how different machine learning models perform when working with imbalanced autonomous vehicle reporting data. Random Forest models provided the strongest overall F1-scores, while Neural Networks showed that improving recall often comes with an increase in false positives.

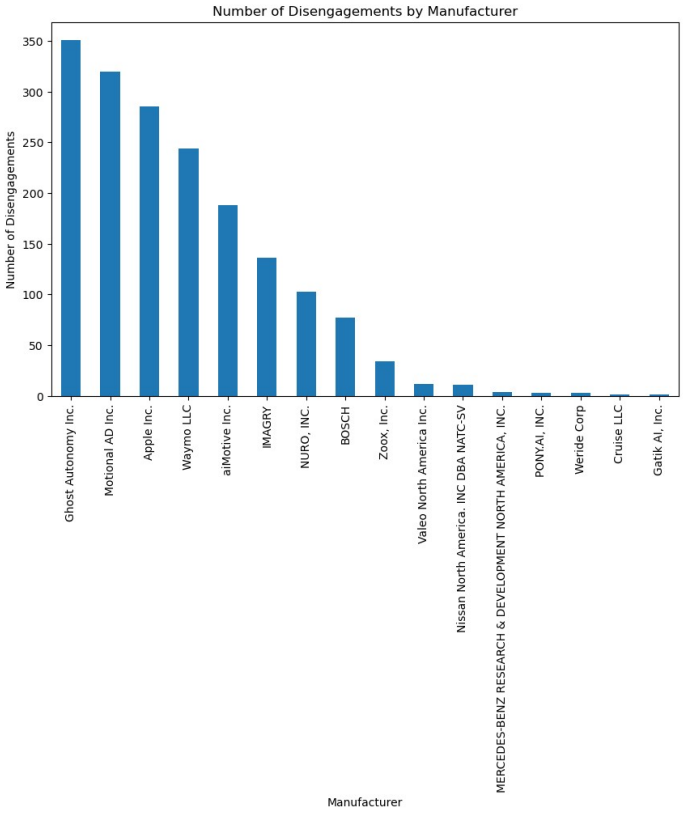
The results also showed me that a higher recall can come with more false positives, and that a model with high accuracy may still miss the collision reports I wanted to identify. This is why I looked at several evaluation metrics instead of relying on accuracy alone.

This project gave me practical experience with data preprocessing, classification models, hyperparameter tuning, and model evaluation. It also helped me understand the importance of interpreting machine learning results carefully rather than relying on a single performance metric.

Appendix A: Additional Visualizations

The following figures come directly from the final Jupyter notebook and provide additional context for the exploratory analysis and model evaluation.

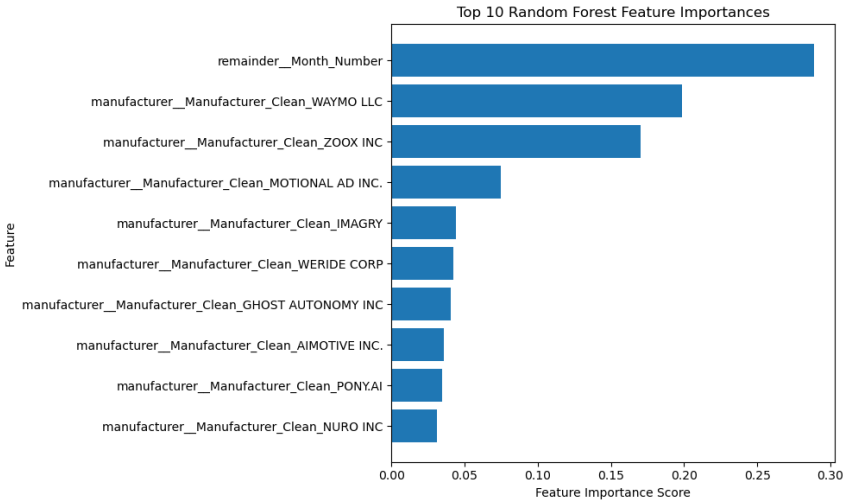
A1. Disengagement Reports by Manufacturer



This chart shows how disengagement report counts vary by manufacturer. Reporting volume alone does not indicate which manufacturer is safer.

Source: Final Jupyter notebook

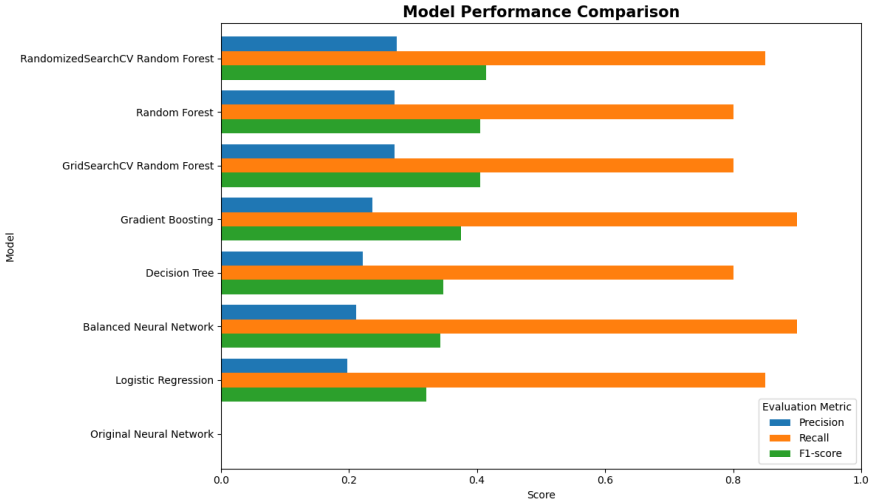
A2. Random Forest Feature Importance



Reporting month and certain manufacturer categories contributed to the model's classification decisions. These importance values describe the model, not the causes of collisions.

Source: Final Jupyter notebook

A3. Model Precision, Recall, and F1-Score



This comparison highlights the trade-off between correctly identifying collision reports and limiting false positives. F1-score is useful because collision reports are less common.

Source: Final Jupyter notebook